Data Analytics & Insights Portfolio

Karen Ogiugo

2026

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SQL

queries.sql +

```sql
-- Create tables
CREATE TABLE CUSTOMERS (
  customer_id INTEGER PRIMARY KEY,
  name TEXT NOT NULL,
  country TEXT NOT NULL
);

CREATE TABLE ORDERS (
  order_id INTEGER PRIMARY KEY,
  customer_id INTEGER NOT NULL,
  order_date DATE NOT NULL,
  amount INTEGER NOT NULL
);

-- Insert data into tables
INSERT INTO CUSTOMERS VALUES (1, 'Karen Ogiugo', 'Ireland');
INSERT INTO CUSTOMERS VALUES (2, 'Sarah Mitchel', 'UK');

INSERT INTO ORDERS VALUES (101, 1, '2024-01-01', 200);
INSERT INTO ORDERS VALUES (102, 1, '2024-02-01', 150);
INSERT INTO ORDERS VALUES (103, 2, '2024-02-10', 300);
```

Figure 1. Create tables

SELECT – retrieving specific columns from a table

```sql
SELECT *
FROM CUSTOMERS;

-- Retrieve specific columns

SELECT name, country
FROM CUSTOMERS;
```

Figure 2. Retrieve columns

Output: 95 ms

customer_id	name	country
1	Karen Ogiugo	Ireland
2	Sarah Mitchel	UK

name	country
Karen Ogiugo	Ireland
Sarah Mitchel	UK

Figure 3. Retrieve columns output

WHERE – filtering data based on conditions

```sql
-- Filter customers from UK

SELECT *
FROM customers
WHERE country = 'UK';

-- Filter orders greater than 200:

SELECT *
FROM orders
WHERE amount > 200;
```

Figure 4. Filter customers

customer_id	name	country
2	Sarah Mitchel	UK

order_id	customer_id	order_date	amount
103	2	2024-02-10	300

Figure 5. Filter customers output

GROUP BY – aggregating data (e.g., totals, averages, counts)

```sql
-- Total amount spent by each customer:

SELECT customer_id, SUM(amount) AS total_spent
FROM orders
GROUP BY customer_id;

-- Count number of orders per customer:

SELECT customer_id, COUNT(order_id) AS total_orders
FROM orders
GROUP BY customer_id;
```

Figure 6. Group data

customer_id	total_spent
1	350
2	300

customer_id	total_orders
1	2
2	1

Figure 7. Group data output

JOIN (**INNER** + **LEFT** – combining data from multiple data tables

```sql
-- INNER JOIN. Show only customers who placed orders:

SELECT c.name, o.order_id, o.amount
FROM customers c
INNER JOIN orders o
  ON c.customer_id = o.customer_id;

-- LEFT JOIN. Shows all customers (even if they have no orders):

SELECT c.name, o.order_id, o.amount
FROM customers c
LEFT JOIN orders o
  ON c.customer_id = o.customer_id;
```

Figure 8. Join data

```
+---------------+----------+--------+
| name          | order_id | amount |
+---------------+----------+--------+
| Karen Ogiugo  |      101 |    200 |
| Karen Ogiugo  |      102 |    150 |
| Sarah Mitchel |      103 |    300 |
+---------------+----------+--------+
+---------------+----------+--------+
| name          | order_id | amount |
+---------------+----------+--------+
| Karen Ogiugo  |      102 |    150 |
| Karen Ogiugo  |      101 |    200 |
| Sarah Mitchel |      103 |    300 |
+---------------+----------+--------+
```

Figure 9. Join data output

BASIC SUBQUERY – a query within another SQL statement.

```sql
-- Customers who spent more than 300 in total:

SELECT name
FROM CUSTOMERS
WHERE customer_id IN (
        SELECT customer_id
        FROM orders
        GROUP BY customer_id
        HAVING SUM(amount) > 300

        );
```

Figure 10. Subquery

```
+---------------+
| name          |
+---------------+
| Karen Ogiugo |
+---------------+
```

Figure 11. Subquery output

Find the customer who spent the most money:

```sql
SELECT c.name, SUM(o.amount) AS total_spent
FROM customers c
JOIN orders o
        ON c.customer_id = o.customer_id
GROUP BY c.name
ORDER BY total_spent DESC
LIMIT 1;
```

Figure 12. Find query

```
+---------------+--------------+
| name          | total_spent |
+---------------+--------------+
| Karen Ogiugo |          350 |
+---------------+--------------+
```

Figure 13. Find query output

Excel

	A	B	C	D	E	F
1	Order ID	Date	Product	Region	Sales	Sales Rep
2	1001	01/01/2024	Laptop	West	1200	Karen Ogiugo
3	1002	02/01/2024	Mouse	East	25	Rachel Flynn
4	1003	02/01/2024	Laptop	West	1300	Karen Ogiugo
5	1004	03/01/2024	Keyboard	East	80	Rachel Flynn

Figure 14. Excel dataset

SORTING DATA – display the highest sales

Figure 15. Sort by largest to smallest

	A	B	C	D	E	F
1	Order ID	Date	Product	Region	Sales	Sales Rep
2	1003	02/01/2024	Laptop	West	1300	Karen Ogiugo
3	1001	01/01/2024	Laptop	West	1200	Karen Ogiugo
4	1004	03/01/2024	Keyboard	East	80	Rachel Flynn
5	1002	02/01/2024	Mouse	East	25	Rachel Flynn

Figure 16. Sort by output

FILTERING DATA – show only "West" region sales

Figure 17. Show only Western sales

	A	B	C	D	E	F
1	Order ID	Date	Product	Region	Sales	Sales Rep
2	1003	02/01/2024	Laptop	West	1300	Karen Ogiugo
3	1001	01/01/2024	Laptop	West	1200	Karen Ogiugo

Figure 18. West filter output

PIVOT TABLES – sum of Sales per product

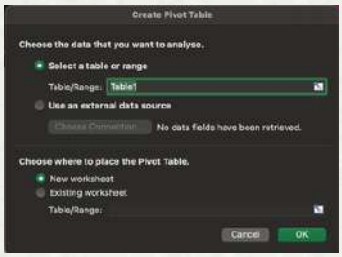

Figure 19. Create pivot table

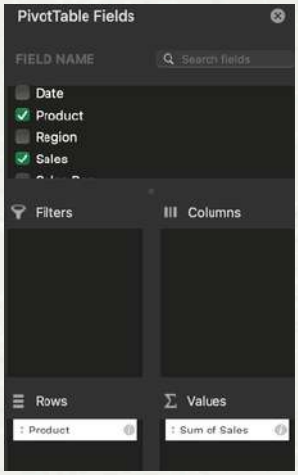

Figure 20. Arrange pivot table fields

	A	B
1		
2		
3	Row Labels	Sum of Sales
4	Keyboard	80
5	Laptop	2500
6	Mouse	25
7	Grand Total	2605

Figure 21. Pivot table output

IF STATEMENTS – label sales as “High” if > 500

	A	B	C	D	E	F	G
1	Order ID	Date	Product	Region	Sales	Sales Rep	Category
2	1003	02/01/2024	Laptop	West	1300	Karen Ogiugo	=IF(E2>500,"High","Low")
3	1001	01/01/2024	Laptop	West	1200	Karen Ogiugo	
4	1004	03/01/2024	Keyboard	East	80	Rachel Flynn	
5	1002	02/01/2024	Mouse	East	25	Rachel Flynn	

Figure 22. IF command

	A	B	C	D	E	F	G
1	Order ID	Date	Product	Region	Sales	Sales Rep	Category
2	1003	02/01/2024	Laptop	West	1300	Karen Ogiugo	High
3	1001	01/01/2024	Laptop	West	1200	Karen Ogiugo	High
4	1004	03/01/2024	Keyboard	East	80	Rachel Flynn	Low
5	1002	02/01/2024	Mouse	East	25	Rachel Flynn	Low

Figure 23. IF command output

VLOOKUP – Bring Category into main table

	A	B	C	D	E	F	G	H
1	Order ID	Date	Product	Region	Sales	Sales Rep	Category	Product Category
2	1003	02/01/2024	Laptop	West	1300	Karen Ogiugo	High	=VLOOKUP(C2, A9:B11, 2, FALSE)
3	1001	01/01/2024	Laptop	West	1200	Karen Ogiugo	High	
4	1004	03/01/2024	Keyboard	East	80	Rachel Flynn	Low	
5	1002	02/01/2024	Mouse	East	25	Rachel Flynn	Low	
6								
7								
8	Product	Category						
9	Laptop	Electronics						
10	Mouse	Accessories						
11	Keyboard	Accessories						
12								

Figure 24. VLOOKUP command

H5 =VLOOKUP(C5, A10:B12, 2, FALSE)

	A	B	C	D	E	F	G	H
1	Order ID	Date	Product	Region	Sales	Sales Rep	Category	Product Category
2	1003	02/01/2024	Laptop	West	1300	Karen Ogiugo	High	Electronics
3	1001	01/01/2024	Laptop	West	1200	Karen Ogiugo	High	Electronics
4	1004	03/01/2024	Keyboard	East	80	Rachel Flynn	Low	Accessories
5	1002	02/01/2024	Mouse	East	25	Rachel Flynn	Low	Accessories
6								
7								
8	Product	Category						
9	Laptop	Electronics						
10	Mouse	Accessories						
11	Keyboard	Accessories						

Figure 25. VLOOKUP command output

XLOOKUP – instead of VLOOKUP

VLOOKUP =XLOOKUP(C2, A9:A11, B9:B11)

	A	B	C	D	E	F	G	H
1	Order ID	Date	Product	Region	Sales	Sales Rep	Category	Product Category
2	1003	02/01/2024	Laptop	West	1300	Karen Ogiugo	High	=XLOOKUP(C2, A9:A11, B9:B11)
3	1001	01/01/2024	Laptop	West	1200	Karen Ogiugo	High	
4	1004	03/01/2024	Keyboard	East	80	Rachel Flynn	Low	
5	1002	02/01/2024	Mouse	East	25	Rachel Flynn	Low	
6								
7								
8	Product	Category						
9	Laptop	Electronics						
10	Mouse	Accessories						
11	Keyboard	Accessories						

Figure 26. XLOOKUP command

H5 =XLOOKUP(C5, A10:A12, B10:B12)

	A	B	C	D	E	F	G	H
1	Order ID	Date	Product	Region	Sales	Sales Rep	Category	Product Category
2	1003	02/01/2024	Laptop	West	1300	Karen Ogiugo	High	Electronics
3	1001	01/01/2024	Laptop	West	1200	Karen Ogiugo	High	Electronics
4	1004	03/01/2024	Keyboard	East	80	Rachel Flynn	Low	Accessories
5	1002	02/01/2024	Mouse	East	25	Rachel Flynn	Low	Accessories
6								
7								
8	Product	Category						
9	Laptop	Electronics						
10	Mouse	Accessories						
11	Keyboard	Accessories						

Figure 27. XLOOKUP command output

CLEANING MESSY DATA – remove extra spaces and fix capitlisation

F	
Sales Rep	C
00 karen Ogiugo	H
00 karen Ogiugo	H
80 rachel Flynn	Lo
25 rachel Flynn	Lo

Figure 28. Data with extra spaces

F	G	H	I
Sales Rep	Category	Product Category	Column1
karen Ogiugo	High	Electronics	=TRIM(F2)
karen Ogiugo	High	Electronics	
rachel Flynn	Low	Accessories	
rachel Flynn	Low	Accessories	

Figure 29. TRIM Command

=TRIM(F3)

E	F	G	H	I
Sales	Sales Rep	Category	Product Category	Column1
1300	karen Ogiugo	High	Electronics	karen Ogiugo
1200	karen Ogiugo	High	Electronics	karen Ogiugo
80	rachel Flynn	Low	Accessories	rachel Flynn
25	rachel Flynn	Low	Accessories	rachel Flynn

Figure 30. TRIM Command output

=PROPER(TRIM(F2))

F	G	H	I	J
Sales Rep	Category	Product Category	Column1	Column2
karen Ogiugo	High	Electronics	karen Ogiugo	=PROPER(TRIM(F2))
karen Ogiugo	High	Electronics	karen Ogiugo	
rachel Flynn	Low	Accessories	rachel Flynn	
rachel Flynn	Low	Accessories	rachel Flynn	

Figure 31. Proper TRIM Command

=PROPER(TRIM(F5))

F	G	H	I	J
Sales Rep	Category	Product Category	Column1	Column2
karen Ogiugo	High	Electronics	karen Ogiugo	Karen Ogiugo
karen Ogiugo	High	Electronics	karen Ogiugo	Karen Ogiugo
rachel Flynn	Low	Accessories	rachel Flynn	Rachel Flynn
rachel Flynn	Low	Accessories	rachel Flynn	Rachel Flynn

Figure 32. Proper TRIM Command output

Tableau

	A	B	C	D	E	F	G	H
1	Order ID	Order Date	City	Product	Category	Quantity	Sales	Cost
2	1	01/01/2024	New York	Latte	Beverage	2	12	6
3	2	01/02/2024	Chicago	Cappuccino	Beverage	1	6	3
4	3	01/03/2024	New York	Croissant	Food	3	15	7
5	4	01/05/2024	Boston	Latte	Beverage	1	6	3
6	5	01/06/2024	Chicago	Muffin	Food	4	16	8
7	6	01/08/2024	Boston	Espresso	Beverage	2	8	4
8	7	01/10/2024	New York	Muffin	Food	2	8	4
9	8	01/12/2024	Chicago	Latte	Beverage	3	18	9
10	9	01/15/2024	Boston	Croissant	Food	2	10	5
11	10	01/18/2024	New York	Espresso	Beverage	4	16	8

Figure 33. Coffee dataset

Imported

Order ID	Order Date	City	Product	Category	Quantity	Sales	Cost
1	01/01/2024	New York	Latte	Beverage	2	12	6
2	01/02/2024	Chicago	Cappuccino	Beverage	1	6	3
3	01/03/2024	New York	Croissant	Food	3	15	7
4	01/05/2024	Boston	Latte	Beverage	1	6	3
5	01/06/2024	Chicago	Muffin	Food	4	16	8
6	01/08/2024	Boston	Espresso	Beverage	2	8	4
7	01/10/2024	New York	Muffin	Food	2	8	4
8	01/12/2024	Chicago	Latte	Beverage	3	18	9
9	01/15/2024	Boston	Croissant	Food	2	10	5
10	01/18/2024	New York	Espresso	Beverage	4	16	8

Figure 34. Coffee dataset imported to Tableau

Bar Chart – total sales by product

Figure 35. total sales by product

Line Chart – sale trend over timet

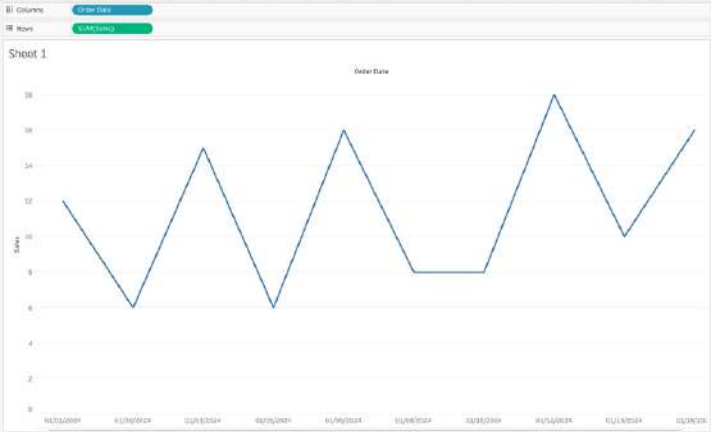

Figure 36. Sale trend over time

Dashboard – sheet 1 and sheet 2

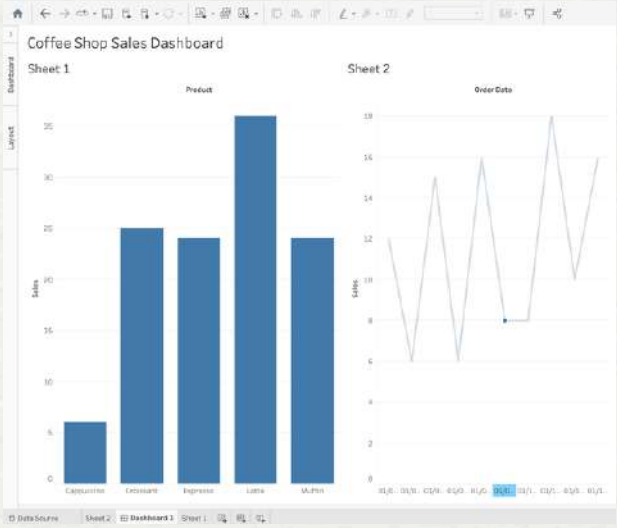

Figure 37. Add to dashboard

Calculated Fields – Example 1

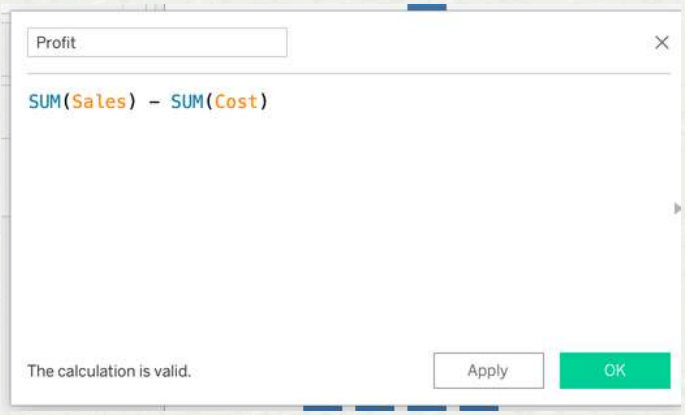

Figure 38. Calculated fields command

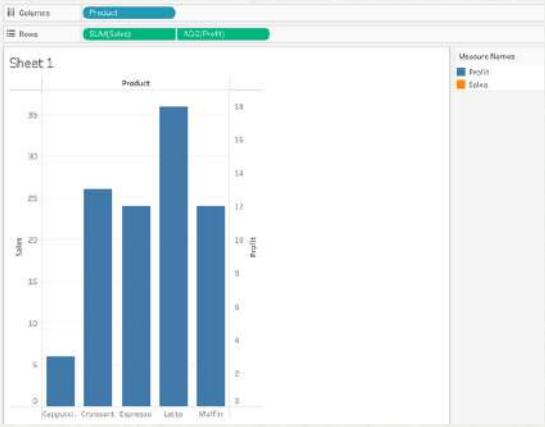

Figure 39. Calculated fields command output

Calculated Fields – Example 2

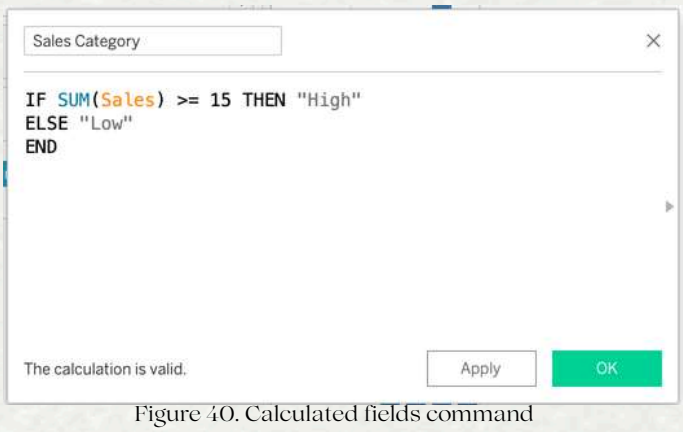

Figure 40. Calculated fields command

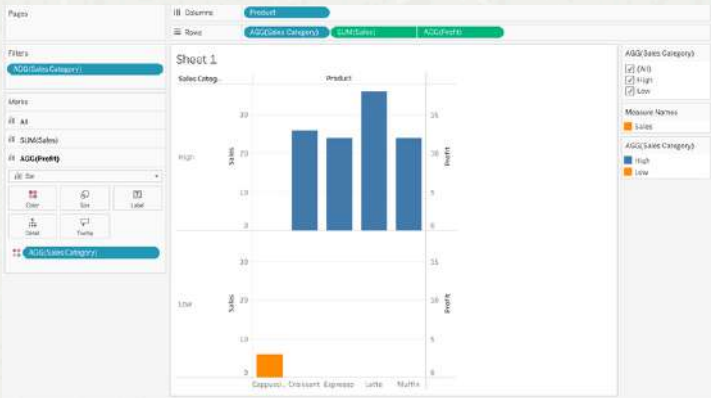

Figure 41. Calculated fields command output

Statistics

Mean – The mean is sensitive to extreme values (outliers).
Use when data is evenly distributed.

Formula: $$\text{Mean} = \frac{\text{Sum of values}}{\text{Number of values}}$$

Example: Sales: 10, 20, 30

Mean = (10 + 20 +30) / 3 = 20

Median – The median is more stable when data has outliers.
Use when data has outliers, for example, salaries or house prices.

Example: Sales: 10, 20, 30, 500

Median = 25

Correlation – Measuring how two variables move together.

- **+1** – Strong positive relationship
- **0** – No relationship
- **-1** – Strong negative relationship

Positive Correlation – As one increases, the other increases.
For example, more advertising leads to more sales.

Negative Correlation – As one increases, the other decreases.
Higher price leads to lower demand.

Correlarion vs Causation – Just because two things move together does not mean one causes the other. Ice cream sales and sunglasses sales increase together. The real cause is summer weather.

Regression Line – The best-fit line through data points, the direction of the relationship and the strength of the trend.

Project 1

Ireland Housing Market Analysis

February 2026

Overview

This project analyses housing market trends in Ireland using SQL and data visualisation. The goal is to understand how property prices vary over time, and whether property size impacts the price.

Data Source: Kaggle

Tools: MySQL Workbench | Excel | Power BI

Figure 42. Neighbourhood in Ireland

Key Questions:

- Which countries have the highest housing prices?
- How have housing prices changed year-over-year?
- Does property size influence price?
- How does Dublin compare to other counties?

Dataset Description

The dataset contains property sales data across Ireland.

Key columns:

- address
- price
- bedrooms
- bathrooms
- property_type
- area
- county
- features
- date_of_construction

Dataset Cleaning – after importing housing data, ensure the analysis uses complete and accurate records.

Check missing values

```
-- Clean the Data

-- Check missing values (by Karen Ogiugo):

SELECT *
FROM housing
WHERE price IS NULL;
```

Figure 43. Find Null values

Remove invalid rows

```
-- Remove invalid rows (by Karen Ogiugo):

DELETE FROM housing
WHERE price IS NULL;
```

Figure 44. Delete Null values

Check counties

```
-- check counties (by Karen Ogiugo):

SELECT DISTINCT county
FROM housing;
```

Figure 45. View counties in dataset

Overall Price Statistics

Overview of the Irish property price range

```
-- market statistics (by Karen Ogiugo)

SELECT
MIN(price) AS lowest_price,
MAX(price) AS highest_price,
AVG(price) AS average_price
FROM housing;
```

Figure 46. Price range

Average Price by County

```
-- Average Price by County (by Karen Ogiugo)

SELECT
county,
AVG(price) AS avg_price
FROM housing
GROUP BY county
ORDER BY avg_price DESC;
```

Figure 47. Average price

Property Type Analysis

```
-- houses vs apartments. (by Karen Ogiugo)

SELECT
property_type,
AVG(price) AS avg_price
FROM housing
GROUP BY property_type
ORDER BY avg_price DESC;
```

Figure 48. Property type

Bedroom vs Price

```
-- how property size affects price. (by Karen Ogiugo)

SELECT
bedrooms,
AVG(price) AS avg_price
FROM housing
GROUP BY bedrooms
ORDER BY bedrooms;
```

Figure 49. Property size

Bathroom vs Price

```
-- Bathrooms vs Price (by Karen Ogiugo)

SELECT
bathrooms,
AVG(price) AS avg_price
FROM housing
GROUP BY bathrooms
ORDER BY bathrooms;
```

Figure 50. Property size

Most Expensive Areas

```
-- Most Expensive Areas (by Karen Ogiugo)

SELECT
area,
AVG(price) AS avg_price
FROM housing
GROUP BY area
ORDER BY avg_price DESC
LIMIT 10;
```

Figure 51. Expensive areas top 10

Housing Age Analysis

```
-- Housing Age Analysis (by Karen Ogiugo)

SELECT
date_of_construction,
AVG(price) AS avg_price
FROM housing
GROUP BY date_of_construction
ORDER BY date_of_construction;
```

Figure 52. Housing age

Dashboard

An interactive dashboard was created using Microsoft Power BI to visually explore housing market trends across Ireland.

Figure 53. Housing analysis dashboard

Average Price by County (Map) – A geographical visualisation of the distribution of property prices across counties in Ireland.

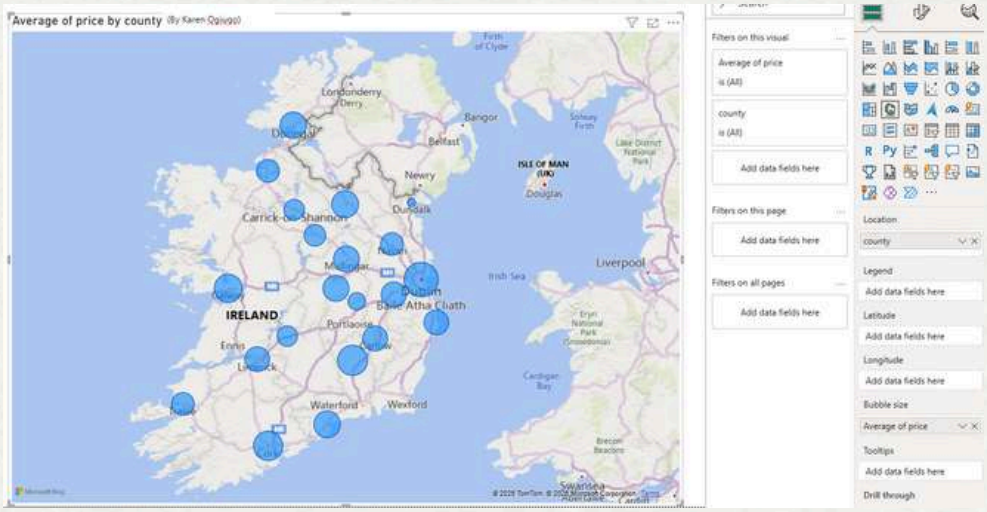

Figure 54. Map

Price vs Bedrooms – This scatter plot shows the relationship between the number of bedrooms and property prices.

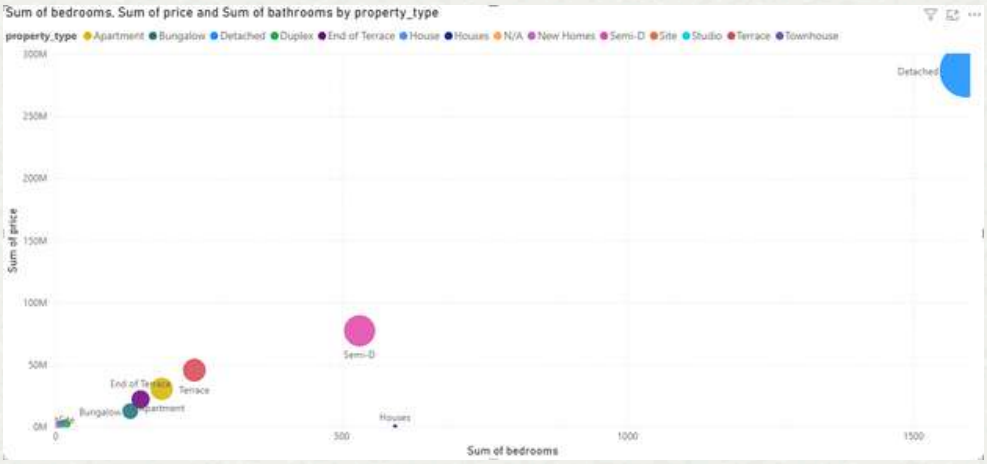

Figure 55. Scatter plot

Price by Property Type– The comparison of the average price across different property types, such as houses and apartments.

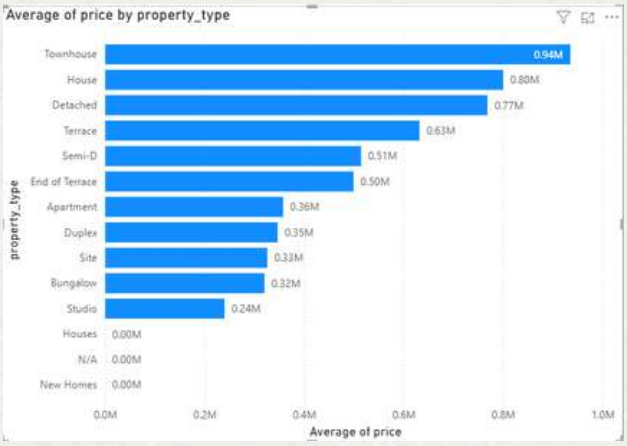

Figure 56. Average price per housing type

Top 10 Most Expensive Areas – Identifying the areas with the highest average property prices, set to top 10.

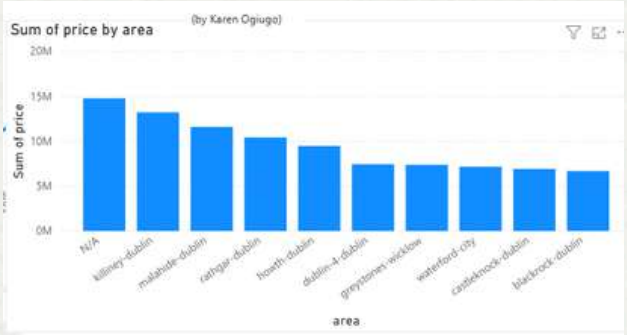

Figure 57. Bar chart

Price Trend by Year – Average Property Price by Construction Year. Newer properties tend to have higher prices

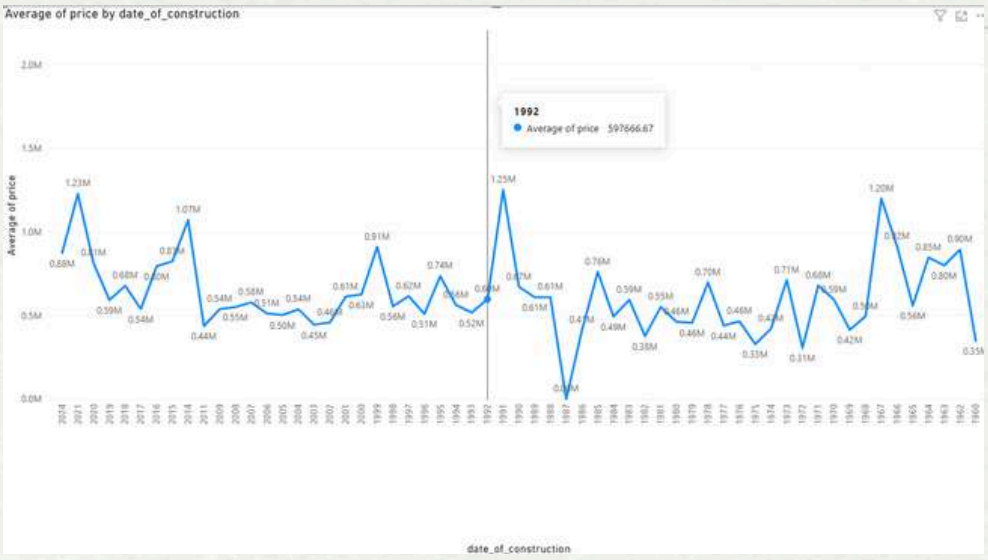

Figure 58. Line chart

Key Insights

Property Location

Properties located in Dublin have the highest average prices compared to other counties. This reflects stronger housing demand and limited supply in the capital region.

Certain local areas show higher average property prices, showing localized housing demand and premium residential homes.

Property Type Impact on Price

Detached houses and larger residential properties tend to have significantly higher prices compared to apartments and smaller property types.

Property Size and Price

Properties with more bedrooms and bathrooms generally require higher prices, indicating that property size is a factor in market value.

Skills Demonstrated

The SQL techniques were used to explore and analyze the housing techniques: **SELECT, WHERE, GROUP BY, AVG, MIN, MAX,** and data cleaning to handle missing values.

Analytical techniques included market trend analysis to understand housing price patterns, geographic analysis to compare housing prices across counties, property segmentation based on property type and property size, followed by exploratory data analysis to identify key housing market insights.

MySQL Workbench was used for data querying and analysis, Microsoft Excel was used for initial data inspection, and Microsoft Power BI for data visualization and dashboard development.

Tesco Ireland Retail Analytics (SQL-Heavy)

March 2026

Overview

This project mimics a retail analytica system for Tesco supermarket operating in Ireland. A relational database was designed to store customer, product, store, and sales data.

With SQL techniques, the analysis explores sales performance, customer purchasing behavior, and product trendds across multiple Irish store locations.

Tools Used: Draw.io, MySQL Workbench, Microsoft Excel, Microsoft Power BI

Figure 59. Tesco store

Project Goal

To understand:

- Which products generate the most revenue
- Which locations perform best
- Customer purchasing behaviour
- Sales trends over time

Relationships

customers orders ›› orders
stores ›› orders
orders ›› order_items
products ›› order_items

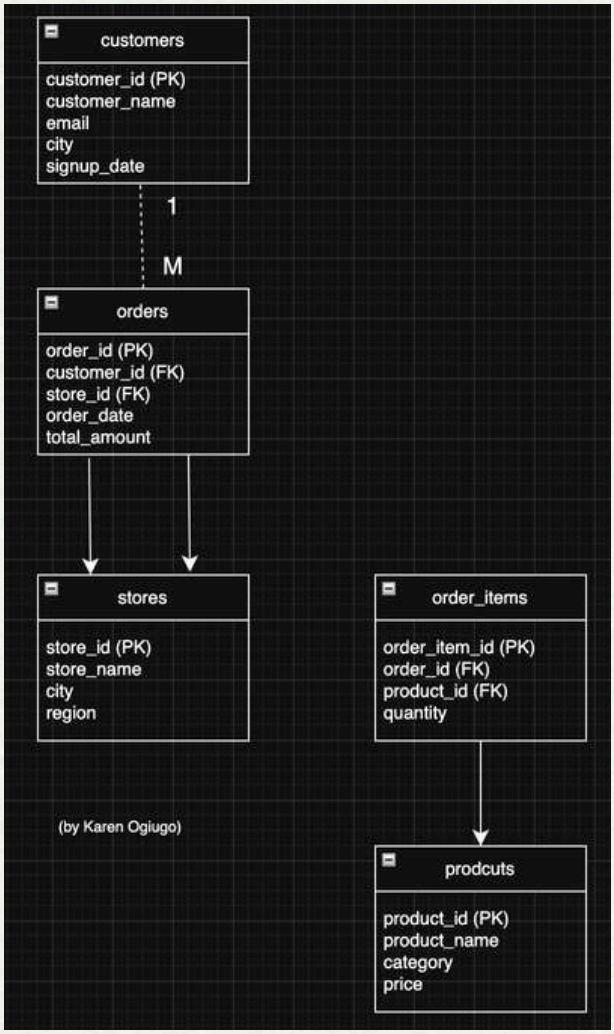

Figure 60. ER diagram

Relationship Explanation

A customer can place multiple orders
Each order is placed at one store.
An order can contain multiple products.
A product can appear in many orders.

Database Creation

The database and tables were created using SQL.
Data was converted from Excel to CSV format, and then imported to MySQL Workbench. This was done for *customers*, *orders*, *stores*, *order_items*, and *products*.

```sql
-- Create Database (by Karen Ogiugo)
CREATE DATABASE tesco_ireland_retail;

USE tesco_ireland_retail;

-- Create Customers Table

CREATE TABLE customers (
    customer_id INT PRIMARY KEY,
    customer_name VARCHAR(100),
    email VARCHAR(100),
    city VARCHAR(50),
    signup_date DATE
);
-- import customers

LOAD DATA LOCAL INFILE
'/Users/karen/Documents/DATA analytics portfolio /tesco customers.csv'
INTO TABLE customers
FIELDS TERMINATED BY ','
ENCLOSED BY '"'
LINES TERMINATED BY '\n'
IGNORE 1 ROWS;
```

Figure 61. Create database, tables, and import dataset

Customers dataset

```sql
-- view dataset
SELECT * FROM customers;
```

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Result Grid Filter Rows: Search Edit: Export/Import:

customer_id	customer_name	email	city	signup_date
1	Jack Smith	jack.smith@email.com	Limerick	2023-04-18
2	Grace Connor	grace.connor@email.com	Cork	2024-07-23
3	Liam O'Brien	liam.o'brien@email.com	Limerick	2023-01-22
4	Ava Connor	ava.connor@email.com	Cork	2023-01-13
5	Noah McCarthy	noah.mccarthy@email.com	Dublin	2023-01-27
6	Emma O'Brien	emma.o'brien@email.com	Waterford	2023-10-22
7	Noah Murphy	noah.murphy@email.com	Dublin	2023-08-03
8	Charlie Kelly	charlie.kelly@email.com	Limerick	2023-07-15
9	Olivia Kelly	olivia.kelly@email.com	Dublin	2024-11-01
10	Amelia O'Brien	amelia.o'brien@email.com	Waterford	2023-08-15

Figure 62. View imorted data

Data Exploration

The initial SQL queries were used to explore dataset facts.

```
-- Data Exploration

-- Count Rows

SELECT COUNT(*) FROM customers;
SELECT COUNT(*) FROM stores;
SELECT COUNT(*) FROM products;
SELECT COUNT(*) FROM orders;
SELECT COUNT(*) FROM order_items;

-- Check Order Date Range

SELECT
MIN(order_date) AS first_order,
MAX(order_date) AS last_order
FROM orders;
```

Figure 63. Count number of rows per dataset, find order date range

These queries help to understand the general size and structure of the imported datasets before performimg analysis, and the range of order dates.

first_order	last_order
2024-01-01	2024-12-31

Figure 64. Date order range

Revenue Analysis

The first business metric calculated was the total sales revnue.

```
SELECT
ROUND(SUM(total_amount),2) AS total_revenue
FROM orders;
```

Figure 65. Sales revenue query

This query calculates the total value of all purchases across the retail system.

total_revenue
8974.00

Figure 66. Sales revenue output

Revenue by Store Location

JOIN was used to combine *orders* and *store* data.

```
-- Revenue by Store Location (JOIN)

SELECT
s.city,
ROUND(SUM(o.total_amount),2) AS total_revenue
FROM orders o
JOIN stores s
ON o.store_id = s.store_id
GROUP BY s.city
ORDER BY total_revenue DESC;
```

Figure 67. Join orders and store location

This query identifiea which cities generate the most retail sales.

city	total_revenue
Waterford	2103.20
Limerick	1804.40
Dublin	1775.00
Galway	1732.60
Cork	1558.80

Figure 68. Join output

Revenue by Product Category

This will identify which product departments drive the most revenue.

```
-- Revenue by Product Category. e.g., Groceries vs Household vs Beverages

SELECT
p.category,
ROUND(SUM(p.price * oi.quantity),2) AS total_revenue
FROM order_items oi
JOIN products p
ON oi.product_id = p.product_id
GROUP BY p.category
ORDER BY total_revenue DESC;
```

Figure 69. Product category revenue

Category results:

category	total_revenue
Dairy	2293.30
Household	1864.20
Meat	1755.00
Groceries	1439.20
Produce	656.10
Beverages	532.40
Bakery	433.80

Figure 70. Product category output

Top Customers

This query identifies high-value customers

```sql
-- Top Customers

SELECT
c.customer_name,
ROUND(SUM(o.total_amount),2) AS total_spent
FROM customers c
JOIN orders o
ON c.customer_id = o.customer_id
GROUP BY c.customer_name
ORDER BY total_spent DESC
LIMIT 10;
```

Figure 71. Join customers and total spent top 10

Retail companies often their marketing strategies on high-spending customers.

High-spending customers:

customer_name	total_spent
Amelia Walsh	492.30
Amelia Connor	361.80
Charlie Kelly	357.30
Emma Doyle	340.20
Olivia Byrne	301.10
Grace Byrne	284.40
Liam Connor	275.00
Emma Smith	270.90
Liam O'Brien	258.40
Grace Murphy	251.70

Figure 72. Join customers and total spent top 10 output

Window Function: allow ranking and trend analysis across datasets.

Ranking custimers by spending:

```sql
-- Rank Top Customers
SELECT
customer_id,
SUM(total_amount) AS total_spent,
RANK() OVER (
ORDER BY SUM(total_amount) DESC
) AS customer_rank
FROM orders
GROUP BY customer_id;
```

Figure 73. Window function

customer_id	total_spent	customer_rank
93	267.40	1
23	225.60	2
100	222.40	3
8	218.90	4
21	204.90	5
70	189.20	6
31	176.00	7
25	172.30	8
98	166.20	9
35	166.10	10
47	165.40	11
91	161.80	12
73	157.50	13
72	145.90	14
77	141.70	15
59	138.40	16
11	137.70	17
10	137.00	18
48	128.60	19
99	120.80	20
75	120.70	21
55	119.70	22
53	119.10	23
16	119.00	24
82	118.60	25
42	118.10	26
65	114.60	27
1	112.80	28
60	109.60	29
26	109.30	30

Figure 74. Window function output

Subquery: Customers spending above average

```sql
-- Subqueries

-- Customers Spending Above Average

SELECT customer_name
FROM customers
WHERE customer_id IN (
    SELECT customer_id
    FROM orders
    GROUP BY customer_id
    HAVING SUM(total_amount) >
    (SELECT AVG(total_amount) FROM orders)
);
```

Figure 75. subquery 1

Subquery:– Products never purchased

```sql
-- Products Never Purchased

SELECT product_name
FROM products
WHERE product_id NOT IN (
    SELECT DISTINCT product_id
    FROM order_items
);
```

Figure 76. subquery 2

Key Insights

Revenue Leaders
Stores located in *Dublin* generated the highest total sales compared to other counties in Ireland.

Product Category Performance
Groceries and beverages, specifcally *Dairy* products, generated the largest share of retail revenue.

Product Demand Gaps
Some products were never purchased, showing potential inventory inefficiencies.

Sales Growth Trends
Revenue analysis showed steady growth over time in retail transactions.

Skills Demonstrated

Relational database design
SQL joins
Subqueries
Window functions
Business Analysis

Project 3

Excel Grocery Sales Dashboard)

March 2026

Overview

This project analyses retail sales performance using Microsoft Excel. The goal is to explore sales trends, identify top-performing products, and understand revenue distribution across different cities and product categories, specifically in Ireland.

Figure 77. Grocery products

Tools Used: Microsoft Excel, Pivot Tables, Slicers, Charts, KPI metrics

The Dataset – 600 rows

	A	B	C	D	E	F	G	H
1	Order_ID	Date	City	Product	Category	Quantity	Price	Revenue
2	1001	2024-06-25	Dublin	Orange Juice	Beverages	6	3.2	19.2
3	1002	2024-03-03	Galway	Bread	Bakery	2	2	4
4	1003	2024-04-13	Galway	Milk	Dairy	6	2.5	15
5	1004	2024-06-10	Dublin	Orange Juice	Beverages	1	3.2	3.2
6	1005	2024-03-07	Galway	Detergent	Household	2	6.5	13
7	1006	2024-02-20	Cork	Croissant	Bakery	6	2.7	16.2
8	1007	2024-05-08	Cork	Croissant	Bakery	5	2.7	13.5
9	1008	2024-06-02	Limerick	Paper Towels	Household	6	4.2	25.2
10	1009	2024-05-08	Limerick	Paper Towels	Household	5	4.2	21
11	1010	2024-03-13	Dublin	Apples	Groceries	3	3	9
12	1011	2024-04-28	Galway	Cola	Beverages	6	2.1	12.6
13	1012	2024-05-09	Cork	Bread	Bakery	1	2	2
14	1013	2024-06-23	Limerick	Croissant	Bakery	4	2.7	10.8
15	1014	2024-01-23	Dublin	Cola	Beverages	6	2.1	12.6
16	1015	2024-01-18	Limerick	Apples	Groceries	4	3	12
17	1016	2024-02-18	Galway	Paper Towels	Household	3	4.2	12.6
18	1017	2024-06-12	Galway	Croissant	Bakery	3	2.7	8.1
19	1018	2024-05-01	Dublin	Milk	Dairy	4	2.5	10
20	1019	2024-01-23	Dublin	Milk	Dairy	1	2.5	2.5
21	1020	2024-03-08	Cork	Cheese	Dairy	5	3.8	19
22	1021	2024-05-25	Galway	Cola	Beverages	6	2.1	12.6
23	1022	2024-02-13	Cork	Milk	Dairy	1	2.5	2.5
24	1023	2024-02-27	Dublin	Pasta	Groceries	3	2.4	7.2
25	1024	2024-04-21	Limerick	Milk	Dairy	1	2.5	2.5
26	1025	2024-01-30	Galway	Milk	Dairy	6	2.5	15
27	1026	2024-03-30	Dublin	Orange Juice	Beverages	3	3.2	9.6
28	1027	2024-05-14	Galway	Pasta	Groceries	4	2.4	9.6
29	1028	2024-02-16	Galway	Croissant	Bakery	3	2.7	8.1
30	1029	2024-04-24	Cork	Cheese	Dairy	5	3.8	19
31	1030	2024-06-12	Dublin	Apples	Groceries	6	3	18
32	1031	2024-05-08	Galway	Orange Juice	Beverages	4	3.2	12.8
33	1032	2024-06-25	Galway	Milk	Dairy	6	2.5	15
34	1033	2024-06-01	Cork	Cola	Beverages	6	2.1	12.6
35	1034	2024-03-14	Limerick	Cola	Beverages	2	2.1	4.2
36	1035	2024-04-10	Galway	Croissant	Bakery	5	2.7	13.5
37	1036	2024-03-03	Dublin	Detergent	Household	3	6.5	19.5
38	1037	2024-01-31	Dublin	Milk	Dairy	2	2.5	5

Figure 78. Grocery dataset

Pivot Table Analysis

Pivot tables were used to summarise the sales data and identify key patterns.

Revenie by City
This analysis compares total sales across different locations.

Revenue by Category
This identifies which product categories generate the most revenue.

Monthly Revenue Trend
Sales were grouped by month to analyse revenue trends over time.

Top Products
Products that generate the most revenue.

Pivot Table 1 - Revenue by City

Row Labels	Sum of Revenue
Cork	1785.9
Dublin	1685.8
Galway	1616.5
Limerick	1711.4
Grand Total	**6799.6**

Pivot Table 2 - Revenue by Category

Row Label	Sum of Revenue
Bakery	1078.3
Beverages	1203.7
Dairy	1298.7
Groceries	1107
Household	2111.9
Grand Total	**6799.6**

Pivot Table 3 - Monthly Revenue

Row Labels	Sum of Revenue
2024-05-08	137.3
2024-04-24	123.1
2024-01-01	111
2024-06-11	97.4
2024-01-23	96.4
2024-04-22	88.6
2024-05-03	85.7
2024-01-21	85.5
2024-06-02	82.7
2024-04-13	82.2
2024-06-12	80.7
2024-01-06	77.8
2024-04-14	76.6

Pivot Table 4 - Top Products

Row Label	Sum of Revenue
Apples	627
Bread	406
Cheese	756.2
Cola	438.9
Croissant	672.3
Detergent	1137.5
Milk	542.5
Orange Juice	764.8
Paper Towels	974.4
Pasta	480
Grand Total	**6799.6**

Figure 79. Grocery pivot tables

Dashboard

An interactive dashboard was created in Excel to visualise the results. The dashboard includes several charts and KPI indicators that allow users to quickly understand the retail sales performance.

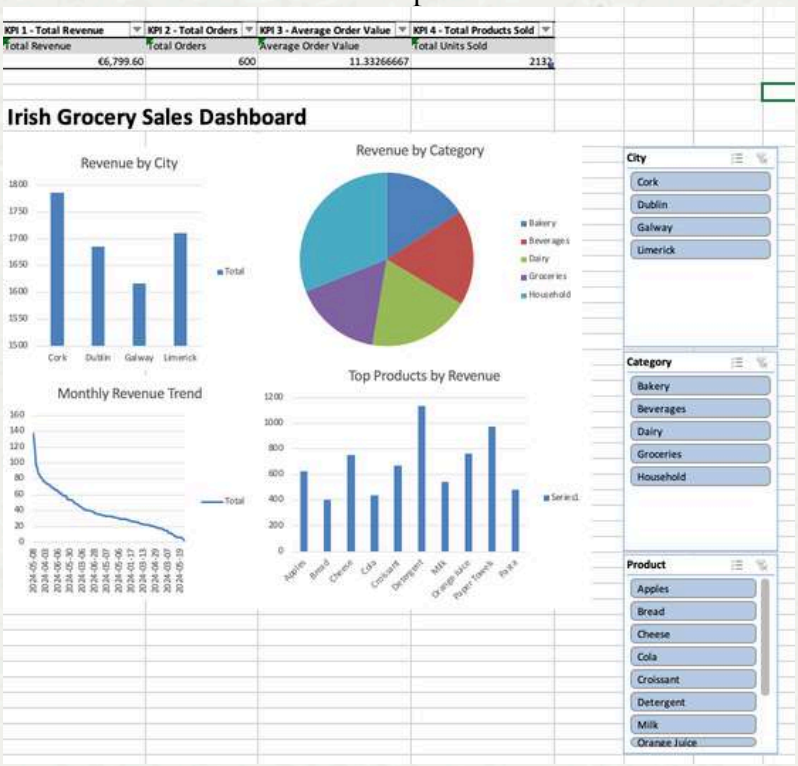

Figure 80. Dashboard with slicers

Dashboard Components

Revenue by City
Column chart showing sales performance across different cities in Ireland.

Revenue by Category
Pie chart displaying the share of revenue generated by each product category.

Monthly Revenue Trend
Line chart illustraing how sales change over time.

Top Products
Bar charts showing the products that bring the most revenue.

KPI Metrics

The dashboard also includes key performance indicators that summarise business performance. The following KPIs allow quick monitoring of overall grocery retail performance:

KPIs:

- Total Revnue
- Total Orders
- Average Order Value
- Total Units Sold

KPI 1 - Total Revenue	KPI 2 - Total Orders	KPI 3 - Average Order Value	KPI 4 - Total Products Sold
Total Revenue	Total Orders	Average Order Value	Total Units Sold
€6,799.60	600	11.33266667	2132

Key Insights

Certain counties in Ireland generate more sales than others.
A few product categories contribute the majority of revenue, such as household goods.
Sales trends show seasonal fluctuations, early summer such as May has the most sales.
A small group of products dives a large portion of total sales. In this case, it is detergent products.

Excel Skills

- Pivot Tables
- Pivot Charts
- KPI calculation
- Slicers and filters
- Interactive dashboard design